PS4 Writeup:

Simulation Without Antibiotic

Fig. 1 shows a simulation of the mean number of bacteria in a patient over a duration of 0 to 300 time steps. There is a log-like growth curve which starts from 0 and ends around 750. Namely at time step 299, we have a population of 765.4 ± 1.4.

At the initial start, population density is low and the bacteria reproduction rate is therefore substantially higher leading to an explosive growth. As the population increase and density increases, reproduction rate decreases - however since the number of bacteria have increased so more reproduction trials occur, the bacteria still monotonically increase. At roughly a timestep of 100, the population plateaus likely meaning the number of reproductions given the population ends up equalling the number of deaths (death rate) and we reach a steady state.

Simulation with Antibiotic

In the simulations with an antibiotic introduced, the initial set of bacteria are set up with no antibiotic resistance. First we look at Simulation 1 highlighted in Fig. 2, with a relatively high birth probability. The bacteria population starts with growth in both total and resistant populations. The total population continues to grow until it reaches a plateau. The resistant population behaves more interestingly, growth slows quickly and decreases after roughly 25 time steps. The explanation for this is that as population increases, the birth rate decreases so deaths quickly catch up, however this happens much faster for resistant bacteria as non-resistant bacteria have a death rate lower by a factor of 4. As a result, the total population is still allowed to climb driven by non-resistant bacteria, leading to the death rate of resistant bacteria exceeding that of its birth rate. Therefore, the non-resistant bacteria end up out-competing the resistant ones, although a stable state for the number of resistant bacteria is reached due to the still ongoing mutations.

The introduction of the antibiotic culls the population of non-resistant bacteria completely, leading to only resistant bacteria reproducing and growing. Then the new plateau is larger than before since birth rates are no longer suppressed by the other form of bacteria, but still smaller than the previous total population plateau as the death rate of resistant bacteria are higher than that of non-resistant (mean total population of 415.8 ± 0.4).

Simulation 2 seen in Fig. 3 is different to Simulation 1 in that the birth rate is reduced. The start is similar with an increase in population, but the plateau is smaller due to the smaller birth rate. This in fact cause a significant behaviour change as resistant bacteria do not decline as in Simulation 1 - the population is kept low so birth rate suppression is no longer significant. The plateau has some slight decrease which may be reflective of the fact population density dependent of the previous time step means the population can slightly spike before it reaches a steady state. Introduction of the antibiotic this time leads to a decay of the bacteria population to 0. This shows the heightened death rate of the resistant bacteria is not sustainable by the lower birth rate and so resistant bacteria growth is mostly driven by mutation rather than reproduction (mean total population of 0.100 ± 0.004).

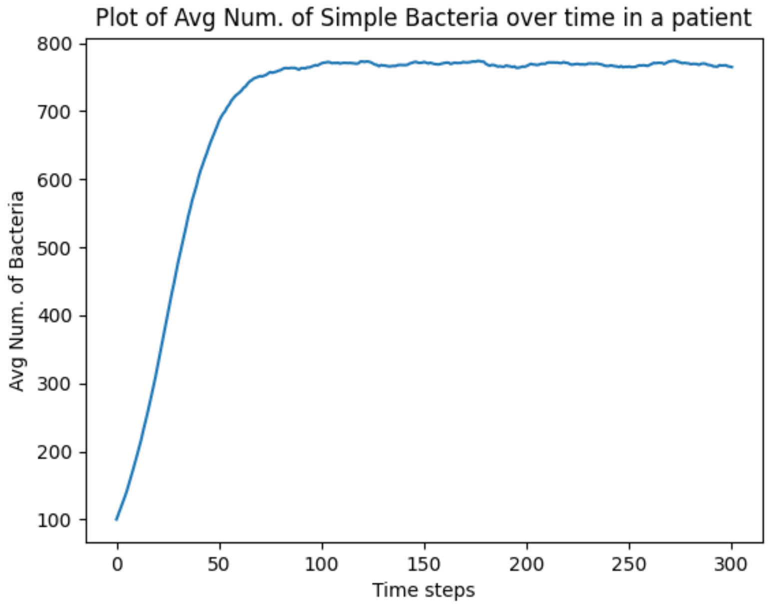

Figure 1: Number of Bacteria = 100, Max Population = 1000, Birth Probability = 0.1, Death Probability = 0.025, Number of Trials = 50

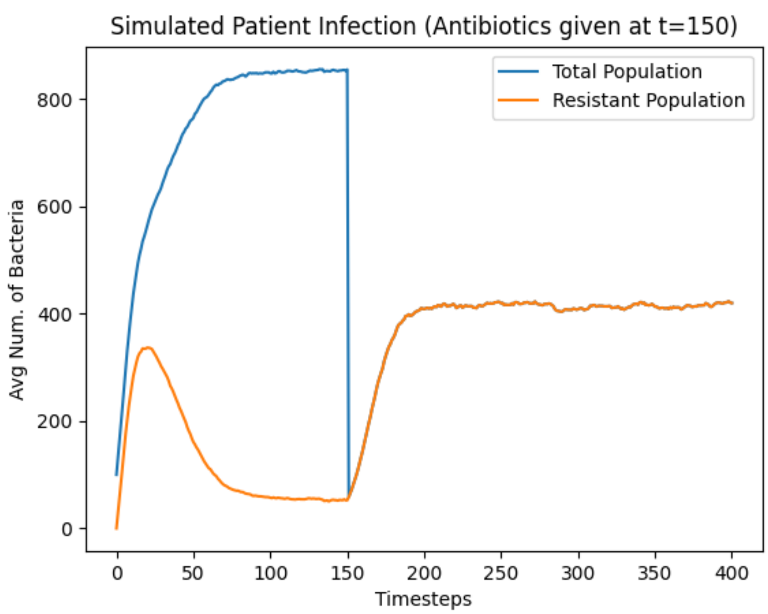

Figure 2: Number of Bacteria = 100, Max Population = 1000, Birth Probability = 0.3, Death Probability = 0.2, Resistant = False, Mutation Probability = 0.8, Number of Trials = 50.

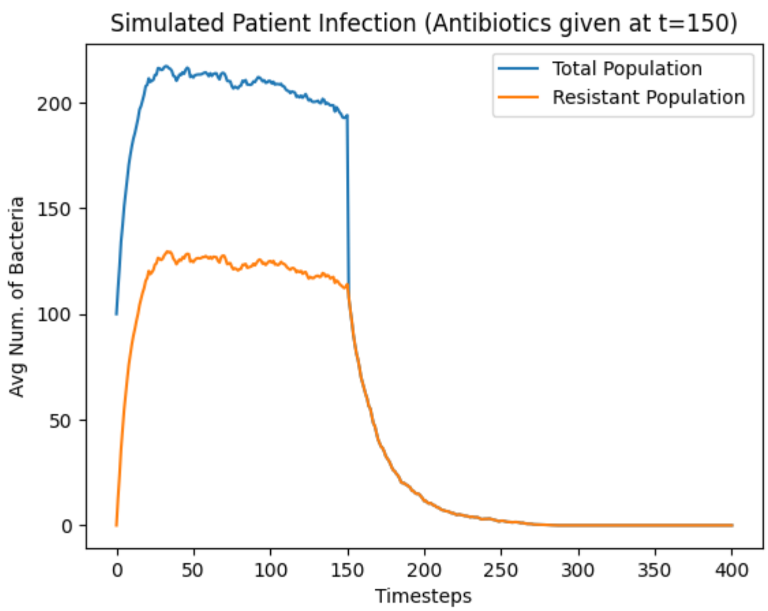

Figure 3: Number of Bacteria = 100, Max Population = 1000, Birth Probability = 0.17, Death Probability = 0.2, Resistant = False, Mutation Probability = 0.8, Number of Trials = 50.